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Marine Survey and Assault Zone Assessment (SAAZA) Course

GySgt J. W. “Chad” Kamphuis





Purpose

- The purpose of this course is to prepare prospective graduates to conduct Survey and Assault Zone Assessment (SAAZA) and formally submit a final product through the use of organic survey equipment, principles of mathematics, and soil evaluation.



Learning Objectives

- Identify the difference between survey and assessment
- Identify publications used to define survey criteria
- Identify airfield categories
- Identify the four phases of a survey
- Identify equipment used to conduct SAAZA



References

- MMT TACSOP
- UFC 3-260-1 Airfield and Heliport Planning and Design
- TM 3-34.48-2 Theater of Operations: Roads, Airfields, and Heliports- Airfield and Heliport Design
- DAFMAN 13-217 Nuclear, Space, Missile, Command and Control; Drop Zone Landing Zone and Helicopter Landing Zone Procedures



References

- Engineering Technical Letter (ETL) 09-6 (17 Aug 09)
- TSPWG 3-260-03.02-19 Tri-Service Pavements Working Group (TSPWG) Manual. Airfield Pavement Evaluation Standards and Procedures



Outline

- Introduction / Overview
- Unit One: Mathematics and Measurement
- Unit Two: Criteria
- Unit Three: Soil Identification and Evaluation
- Unit Four: SAAZA Process
- Unit Five: SAAZA Submission



Outline

- Introduction
- Unit One: Mathematics and Measurement
- Unit Two: Criteria
- Unit Three: Soil Identification and Evaluation
- Unit Four: SAAZA Process
- Unit Five: SAAZA Submission



Outline

- Introduction
 - Definitions
 - Publications
 - Airfield Categories
 - Phases
 - Equipment



Outline

- Introduction
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 - Equipment



Definitions

- Survey
 - To accurately determine area, contour, or boundaries by measuring lines and angles according to the principles of geometry and trigonometry
 - Raw, factual data regarding a location

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Definitions

- Assessment
 - To determine worth or condition
 - This is in relation to published standards and criteria
 - For MMT purposes a focus on flight safety
 - Supported by the accuracy of the survey

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Definitions

- Hasty
 - To provide maximum usable information within a compressed timeline
 - For survey, the prioritization of essential information
 - For assessment, focus remains on minimum criteria

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Outline

- Introduction / Overview
 - Definitions
 - Publications
 - Airfield Categories
 - Phases
 - Equipment



Publications

- UFC 3-260-1
 - Basis for all airfield design criteria
 - Chapter 7 speaks specifically to KC-130/C-17 Semi-prepared landing zones
- TM 3-34.48-2
 - US Air Force (USAF) source for airfield design criteria
- DAFMAN 13-217 (CCT TACSOP)
 - USAF guiding document for landing zone and drop zone procedures
 - Prescribes the procedures, techniques and requirements for operating AZs
 - Governs the content, documentation and approval process for USAF AZ surveys
 - Uses criteria found in TM 3-34.48-2, ETL 09-6, and TSPWG 3-260-03.02-19



Publications

- Engineering Technical Letter 09-6
 - Supersedes criteria contained in AFJPAM 32-8013 Vol 2, Chapter 11, “Airfield Design”
 - Dimensional criteria and guidance for planning, design, construction and evaluation of AZs for C-17 and C-130 aircraft
- MMT TACSOP
 - Based on DAFMAN 13-217
 - MMT guiding document for landing zone and drop zone procedures
 - Prescribes the procedures, techniques and requirements for operating AZs
 - Governs the content, documentation and approval process for USMC AZ surveys



Publications

- TSPWG 3-260-03.02-19 Tri-Service Pavements Working Group (TSPWG) Manual. Airfield Pavement Evaluation Standards and Procedures
 - Supersedes guidance contained in Engineering Technical Letter 02-19
 - Provides criteria and guidance for the structural evaluation of airfields using conventional methods



Outline

- Introduction / Overview
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 - Publications
 - Airfield Categories
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 - Equipment



Airfield Categories

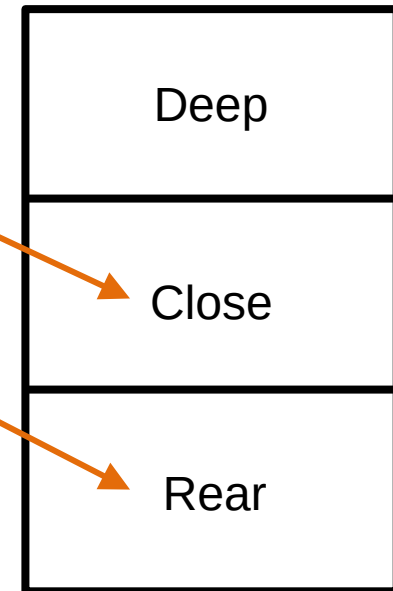
- Initial Airfields
 - Drop zone
 - Extraction zone
 - Expedient airfield
 - Close battle area airfield
 - Support area airfield

32-8013, Vol II; 10-5



Airfield Categories

- Initial Airfields
 - Expedient airfield
 - Close battle area airfield
 - Support area airfield

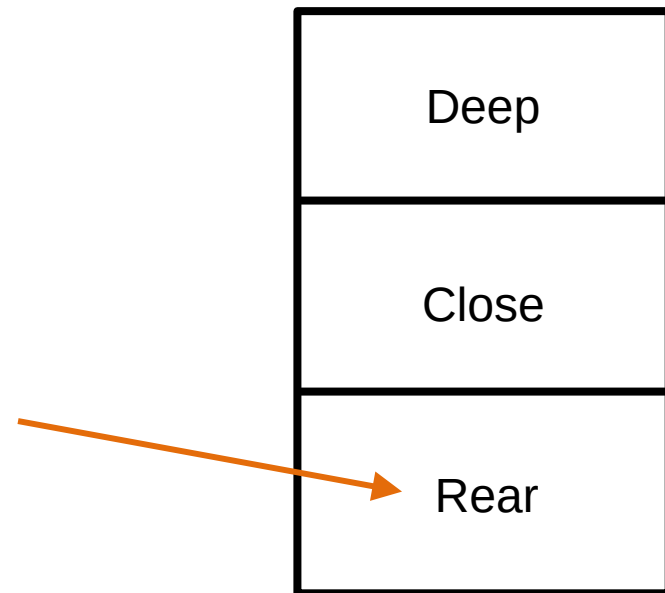


32-8013, Vol II; 10-5



Airfield Categories

- Temporary airfield
 - Assault Landing Zone

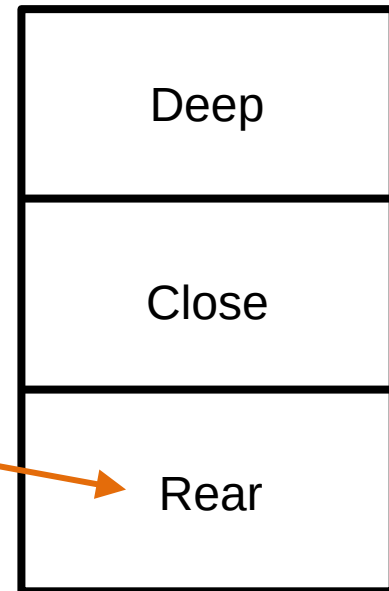


32-8013, Vol II; 10-5



Airfield Categories

- Semi-permanent airfield
 - Airport / Main Air Base



32-8013, Vol II; 10-5



Outline

- Introduction / Overview
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Phases

1. Information Gathering
2. Preparation
3. Site Visit
4. Final Product Development

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Outline

- Introduction / Overview
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Equipment

- Laser Range Finder (LRF)
- Dynamic Cone Penetrometer (DCP)
- Clinometer
- Defense Advanced GPS Receiver (DAGR)
- Line Level
- Measuring Wheel
- Compass
- Digital Camera



Review

- Identify the difference between survey and assessment
 - Survey - To accurately determine area, contour, or boundaries by measuring lines and angles according to the principles of geometry and trigonometry
 - Assessment - To determine worth or condition in relation to published standards and criteria with a focus on flight safety



Review

- Identify publications used to define survey criteria
- UFC 3-260-1
 - Chapter 7 speaks specifically to KC-130/C-17 Semi-prepared landing zones
- TM 3-34.48-2 Theater of Operations: Roads, Airfields, and Heliports- Airfield and Heliport Design
 - US Air Force (USAF) source for airfield design criteria
- Engineering Technical Letter 09-6
 - Dimensional criteria
- TSPWG 3-260-03.02-19 Tri-Service Pavements Working Group (TSPWG) Manual. Airfield Pavement Evaluation Standards and Procedures
 - Structural Criteria



Review

- Identify airfield categories
 - Initial airfields
 - Temporary airfields
 - Semi-permanent airfields



Review

- Identify the four phases of a survey
 1. Information gathering
 2. Preparation
 3. Site Visit
 4. Final product development



Review

- Identify equipment used to conduct SAAZA
 - Laser Range Finder
 - Dynamic Cone Penetrometer
 - Clinometer
 - Defense Advanced GPS Receiver (DAGR)
 - Line Level
 - Measuring Wheel
 - Compass
 - Digital Camera



10 MIN BREAK



Outline

- Introduction / Overview
- **Unit One: Mathematics and Measurement**
- Unit Two: Landing Zone Criteria
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SAAZA Unit 1: Mathematics and Measurement

GySgt J. W. "Chad" Kamphuis





Purpose

- The purpose of this course is to instruct prospective graduates on the proper use of mathematic principles to support the execution of Survey and Assault Zone Assessment (SAAZA).



Learning Objectives

- Determine angles, distances, and horizontal ranges with trigonometric functions
- Identify units of measurement used when conducting SAAZA
- Convert horizontal units of measurement



References

- MMT TACSOP
- UFC 3-260-1 Airfield and Heliport Planning and Design
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Outline

- Mathematics
- Measurement
- Review
- Summary



Outline

- Mathematics
- Measurement
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Outline

- Mathematics
 - Trigonometry
 - Gradient / Glideslope
- Measurement
- Review
- Summary



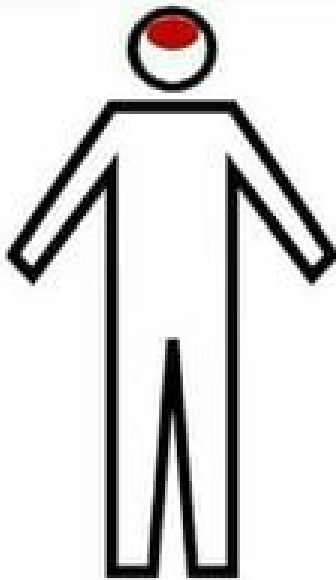
Outline

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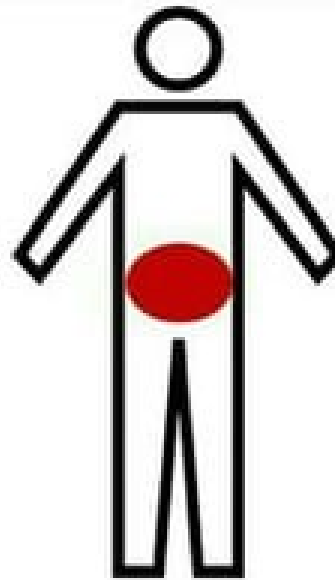


Trigonometry

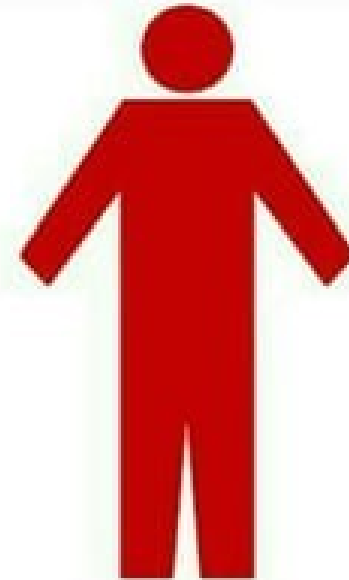
Where does it hurt?



headache



stomach pain

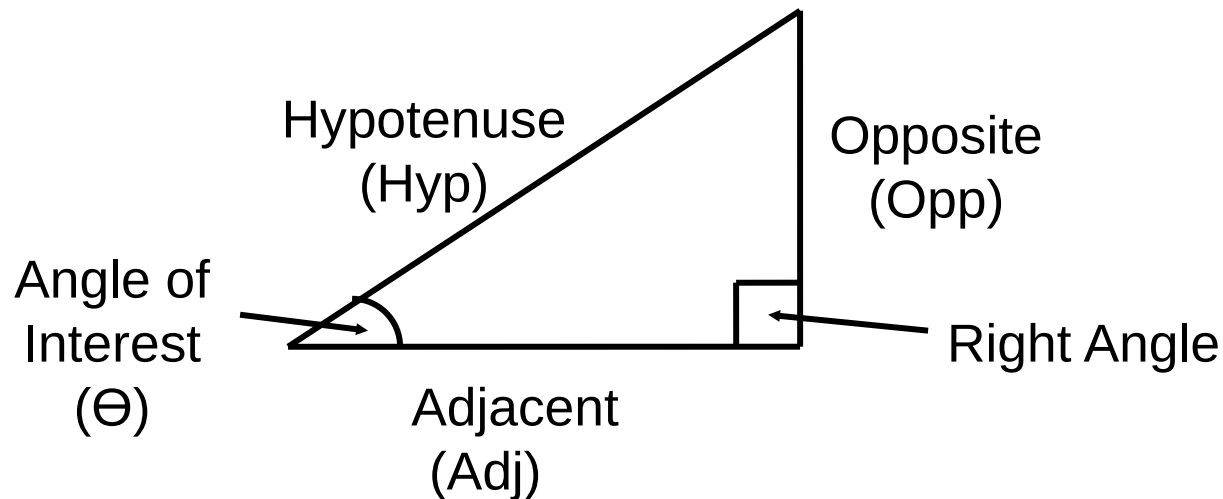


trigonometry



Trigonometry

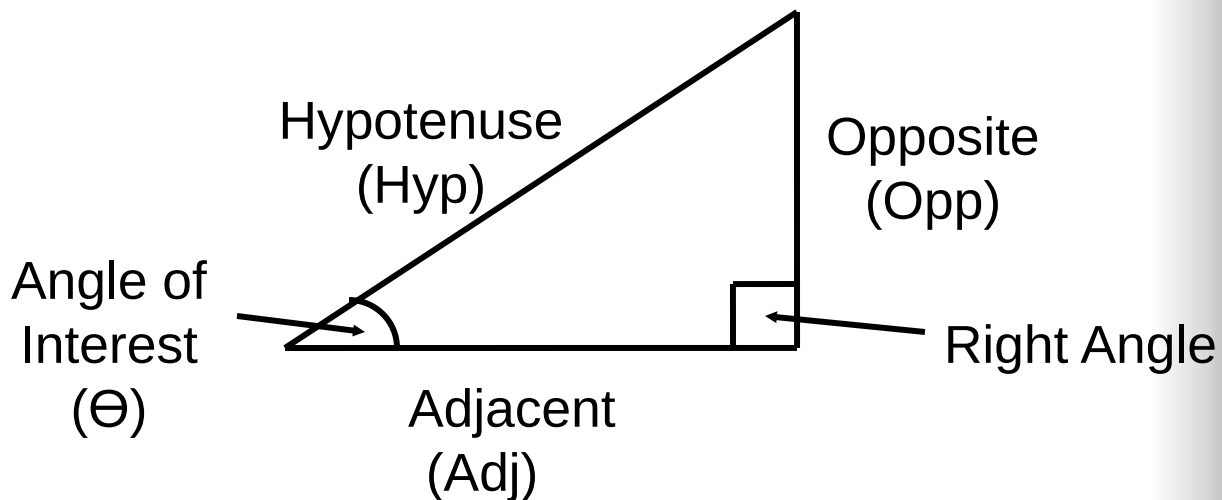
- Surveying = right triangles





Triangles

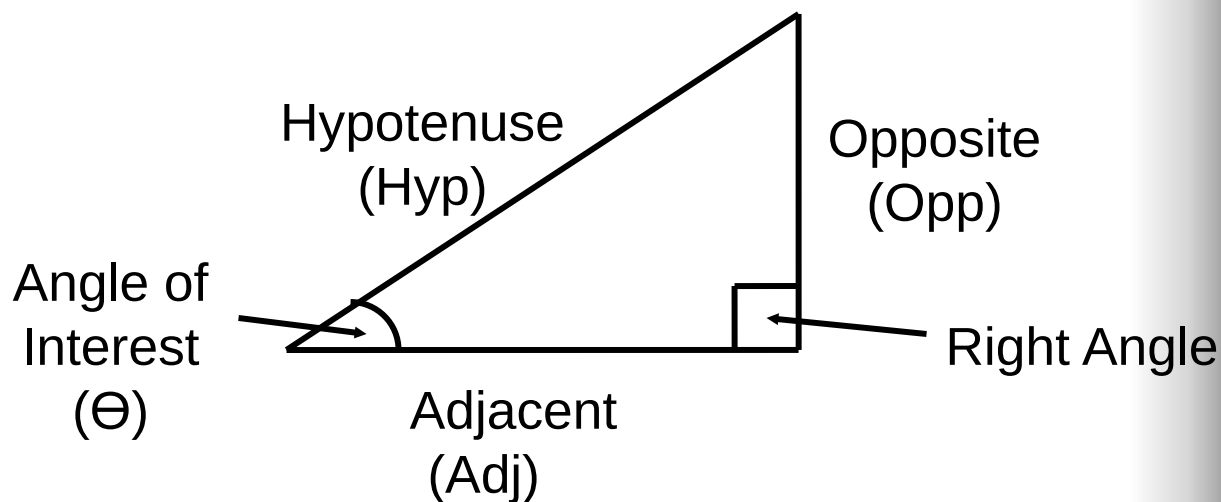
- Solving for Θ
 - Θ is derived from a ratio
 - Opposite or adjacent over the hypotenuse
 - Opposite over the adjacent





Triangles

- Solving for Θ
 - Defined by what sides were used to determine it
 - $\text{Opp} / \text{Hyp} = \text{Sine ratio (Sin)}$
 - $\text{Adj} / \text{Hyp} = \text{Cosine ratio (Cos)}$
 - $\text{Opp} / \text{Adj} = \text{Tangent ratio (Tan)}$



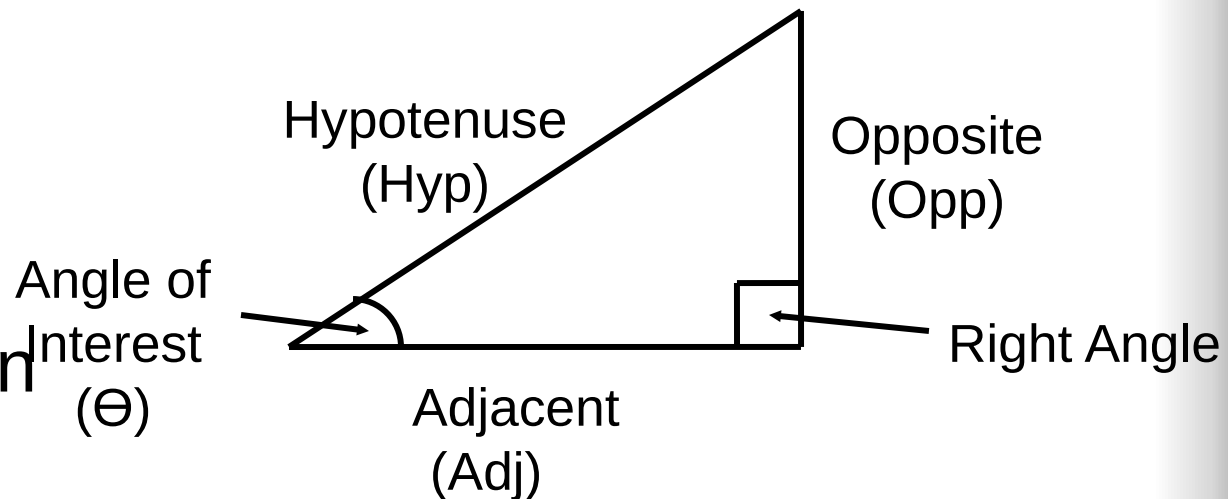


Triangles

- Solving for Θ

- These formulas have only solved for a ratio

- Sin, Cos, Tan represents the AI

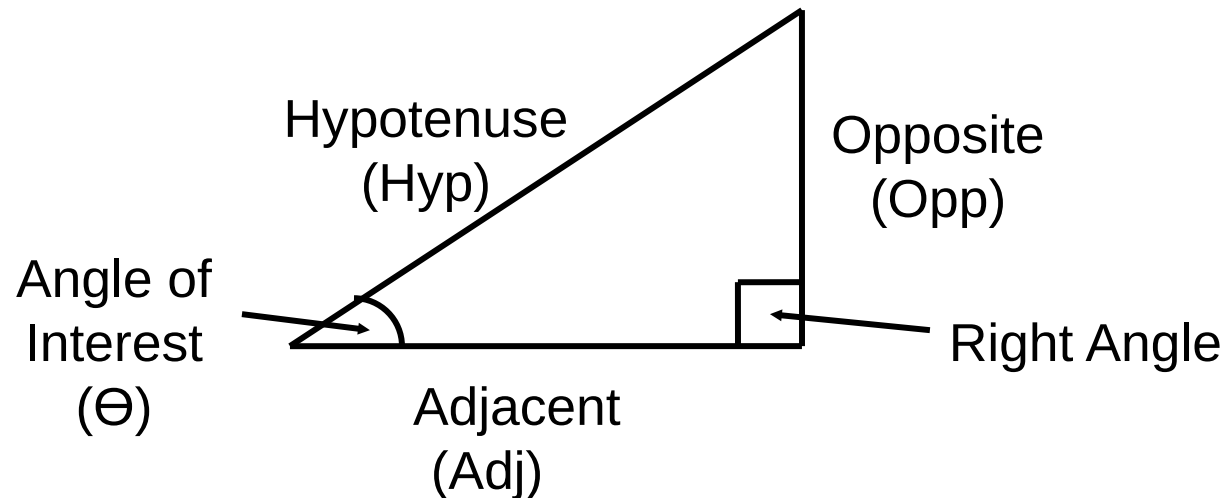


$$\begin{array}{c} \text{Opp} \\ \hline \text{Sin} \mid \text{Hyp} \end{array} \quad \begin{array}{c} \text{Adj} \\ \hline \text{Cos} \mid \text{Hyp} \end{array} \quad \begin{array}{c} \text{Opp} \\ \hline \text{Tan} \mid \text{Adj} \end{array}$$



Trigonometry

- Solving for Θ
 - A problem always provides two Components

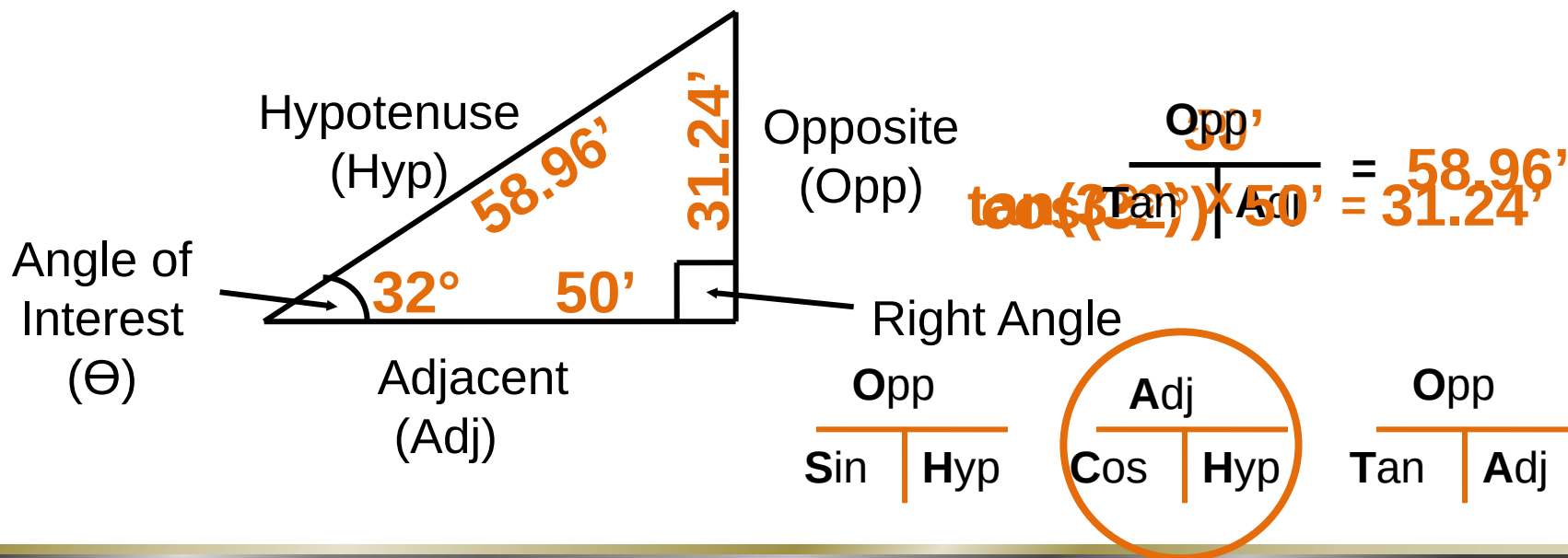


$$\begin{array}{ccc} \frac{\text{Opp}}{\text{Hyp}} & \frac{\text{Adj}}{\text{Hyp}} & \frac{\text{Opp}}{\text{Adj}} \\ \text{Sin} & \text{Cos} & \text{Tan} \end{array}$$



Trigonometry

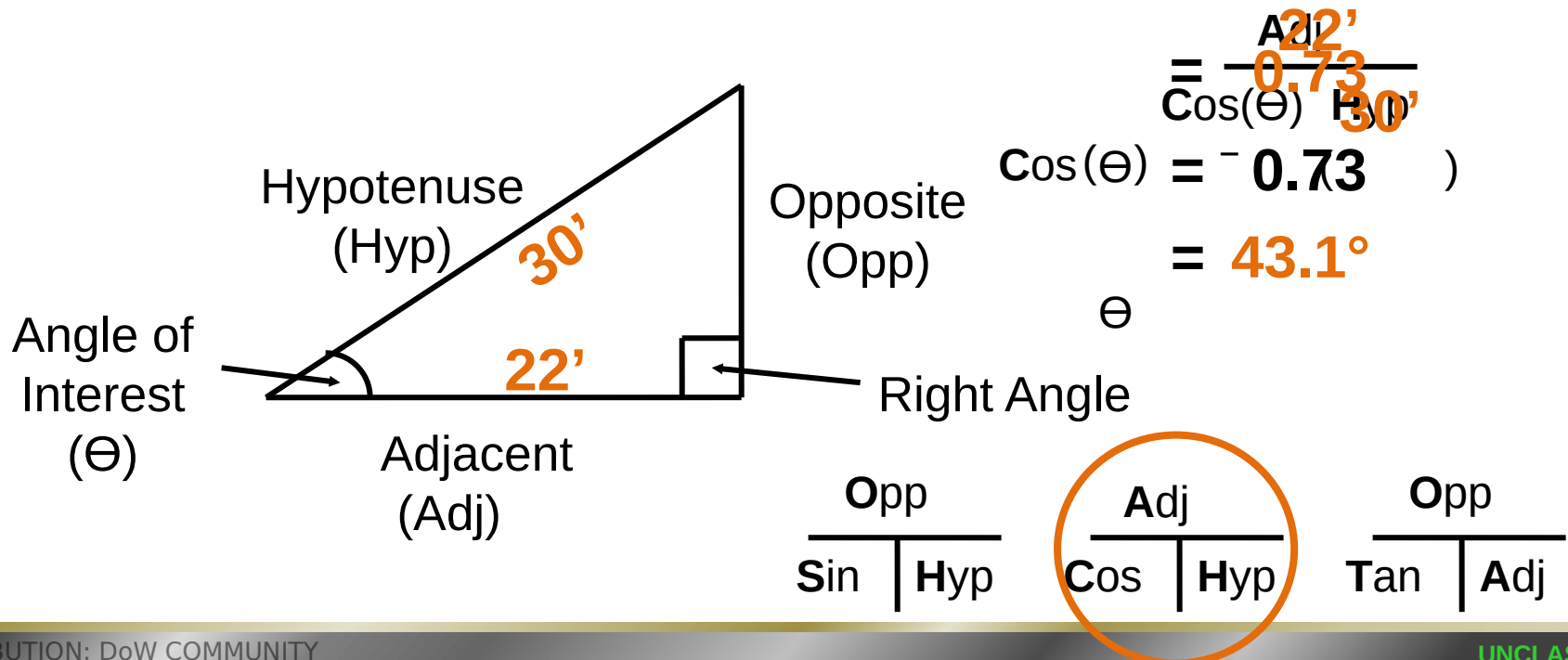
- Solving for Hyp / Adj / Opp
 - To decide what formula to use
 - If a component provided is in the top row, divide it by the component inserted into the bottom row
 - If both components are in the bottom row, multiply





Trigonometry

- Solving for Θ
 - To decide what formula to use, ask yourself
 - If the angle is unknown, you must convert the ratio into an angle using the inverse function.





Trigonometry

- Example 1
 - The height of a pole is known to be 30'
 - The distance of the pole from your position is 75'
 - What is the angle to the top of the pole?
- First: Identify the formula

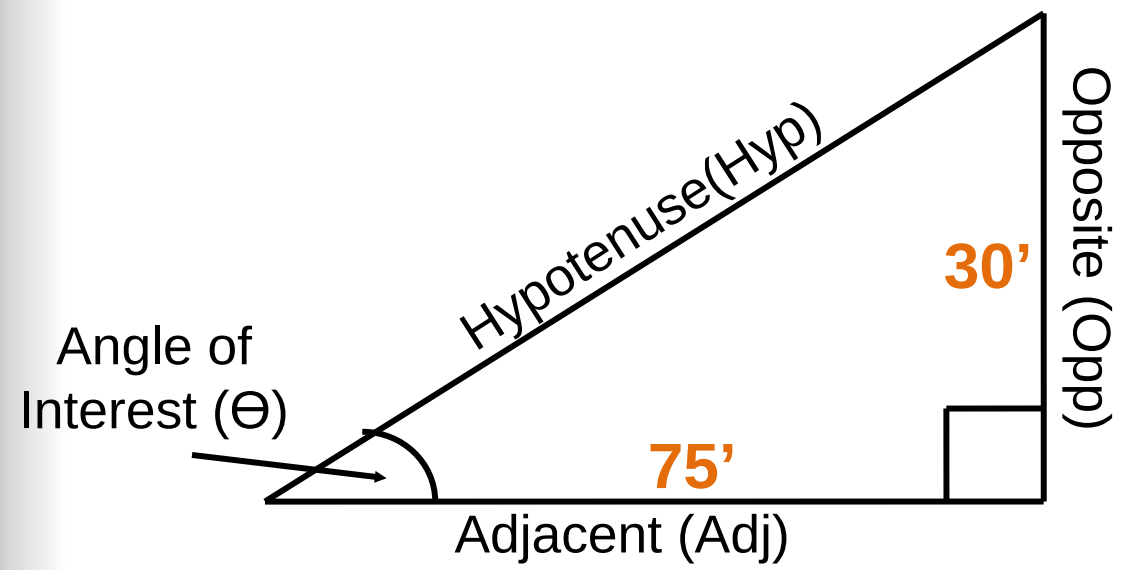
$$\frac{30'}{S | H}$$

$$\frac{75'}{C | H}$$

$$\frac{30'}{T | 75'}$$



Trigonometry



$$\tan(\Theta) = 30 \div 75$$

$$\tan(\Theta) = .4$$

$$\Theta = \tan^{-1}(.4)$$

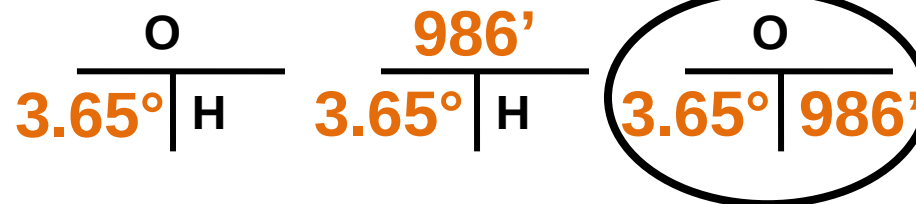
$$\Theta = 21.8^\circ$$

30'
$\tan$ 75'



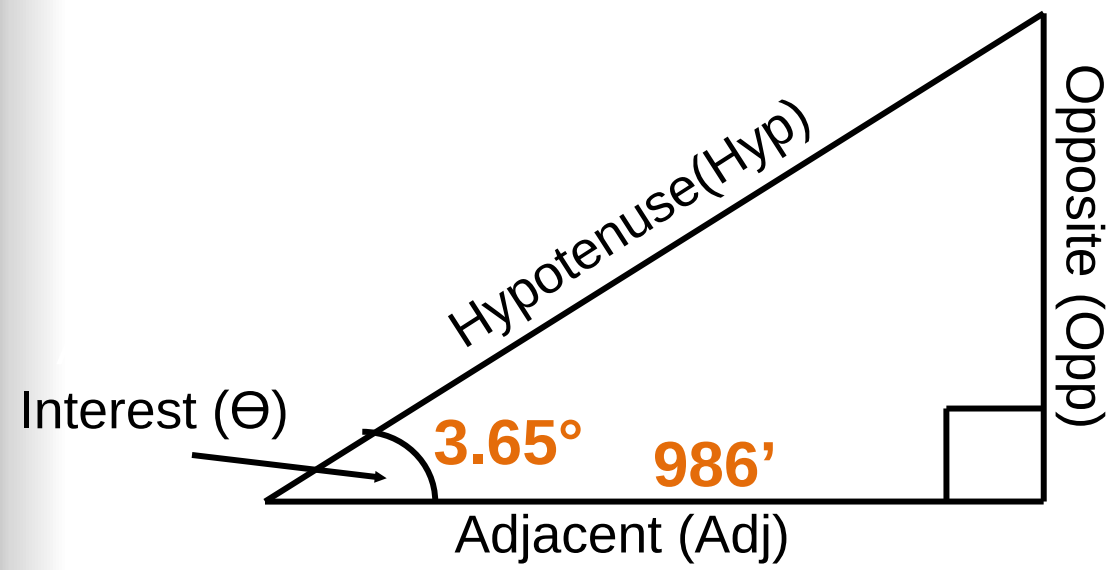
Trigonometry

- Example 2
 - The angle to the top of a tower is 3.65°
 - Distance to the tower is 986'
 - Find the Height of the Tower





Trigonometry



$$\begin{aligned}\tan(3.65) \times 986 &= \text{Opp} \\ .063 \times 986 &= \text{Opp} \\ 62.9 &= \text{Opp}\end{aligned}$$

+  observer height

$$\begin{array}{r} 0 \\ \hline 3.65^\circ \mid 986' \end{array}$$



Trigonometry

Instructor: Great job. Did you add the observer height?

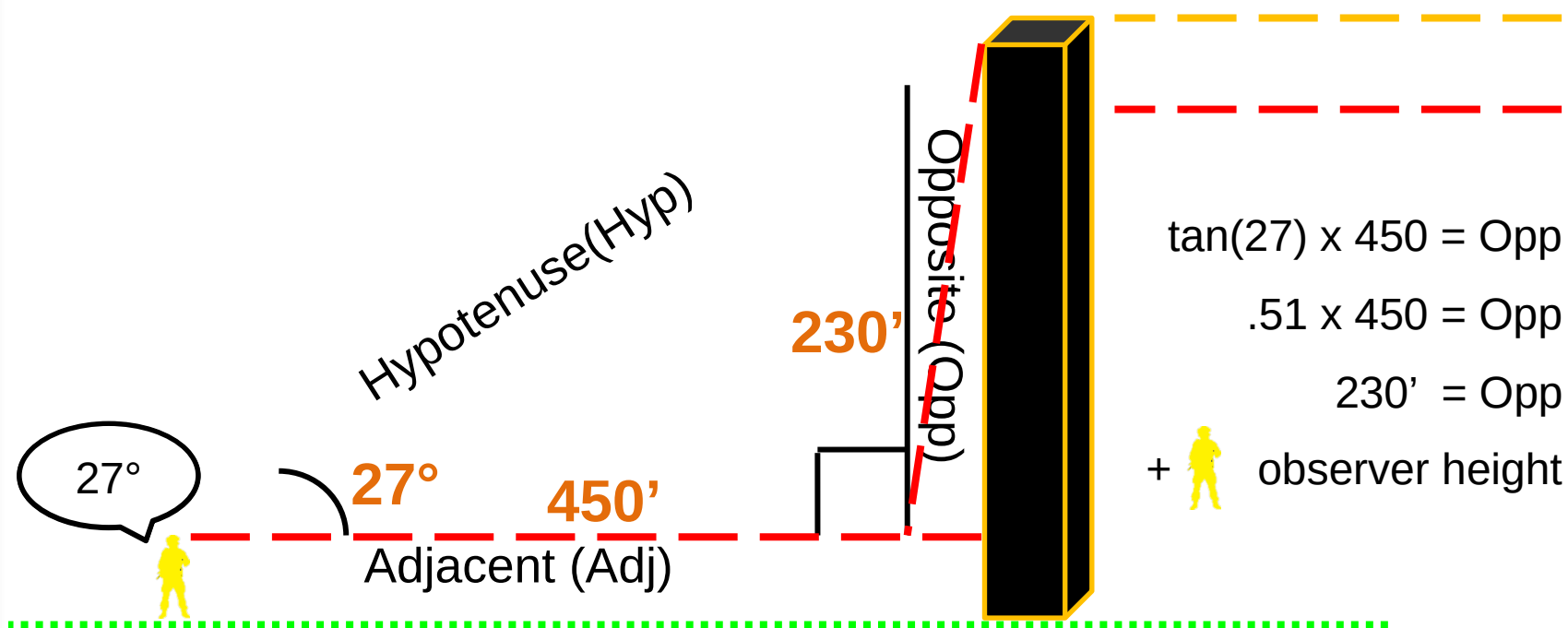
Students:





Trigonometry

- Observer height

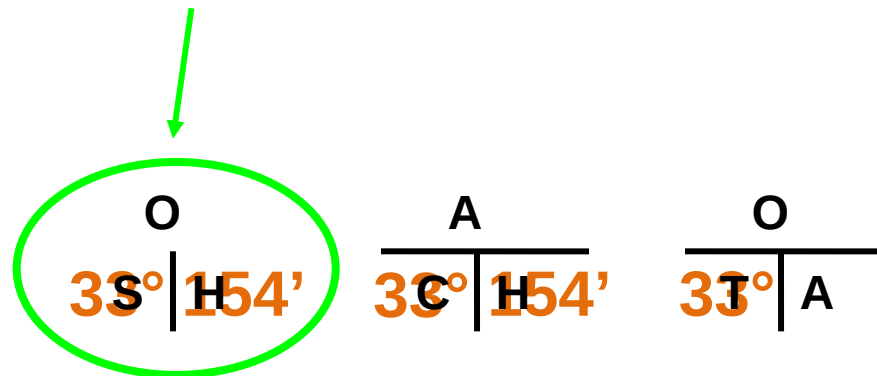




Trigonometry

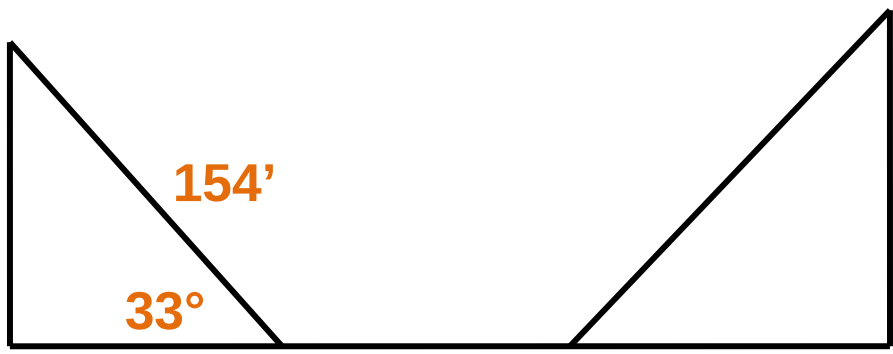
• Example 3

- Angle to the top of a seating section in a football stadium is 33°
- Straight-line distance from the field to the top of the seating section is 154'
- Find the height of the seating section from the ground of the stadium





Trigonometry

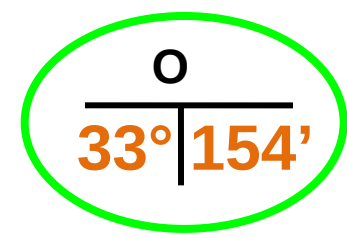


$$\sin(33) \times 154 = \text{Opp}$$

$$.54 \times 154 = \text{Opp}$$

$$83.9 = \text{Opp}$$

+  observer height





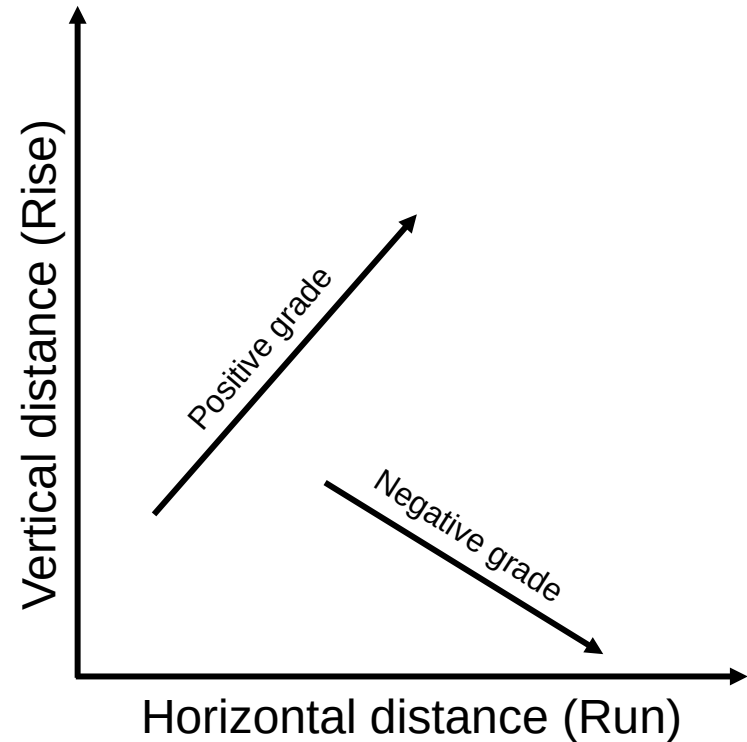
Outline

- Mathematics
 - Trigonometry
 - Gradient / Glideslope
- Measurement
- Review
- Summary



Gradient

- Definitions
 - Positive grade (+)
 - Increase in elevation
 - Negative grade (-)
 - Decrease in elevation
 - Rise
 - Vertical distance (elevation)
 - Run
 - Horizontal distance
 - Decimal grade
 - Rise divided by run
 - Percent grade
 - Decimal grade multiplied by 100
 - What we use in survey





Decimal / Percent Grade

- Example:
 - 100' down the taxiway there is a 2.5' positive rise in elevation

$$\begin{array}{l} \text{Rise} \\ \text{Run} \end{array} \quad \frac{2.5'}{100'} = +.025 \text{ decimal grade} \quad \times 100 = +2.5\% \text{ grade}$$



Decimal / Percent Grade

- Example:
 - 200' down the taxiway there is a 1.3' NEGATIVE rise in elevation

$$\begin{array}{l} \text{Rise} \\ \text{Run} \end{array} \quad \frac{1.3'}{200'} = -.0065 \text{ decimal grade} \quad \times 100 = -.65\% \text{ grade}$$



Recording Gradient Methods

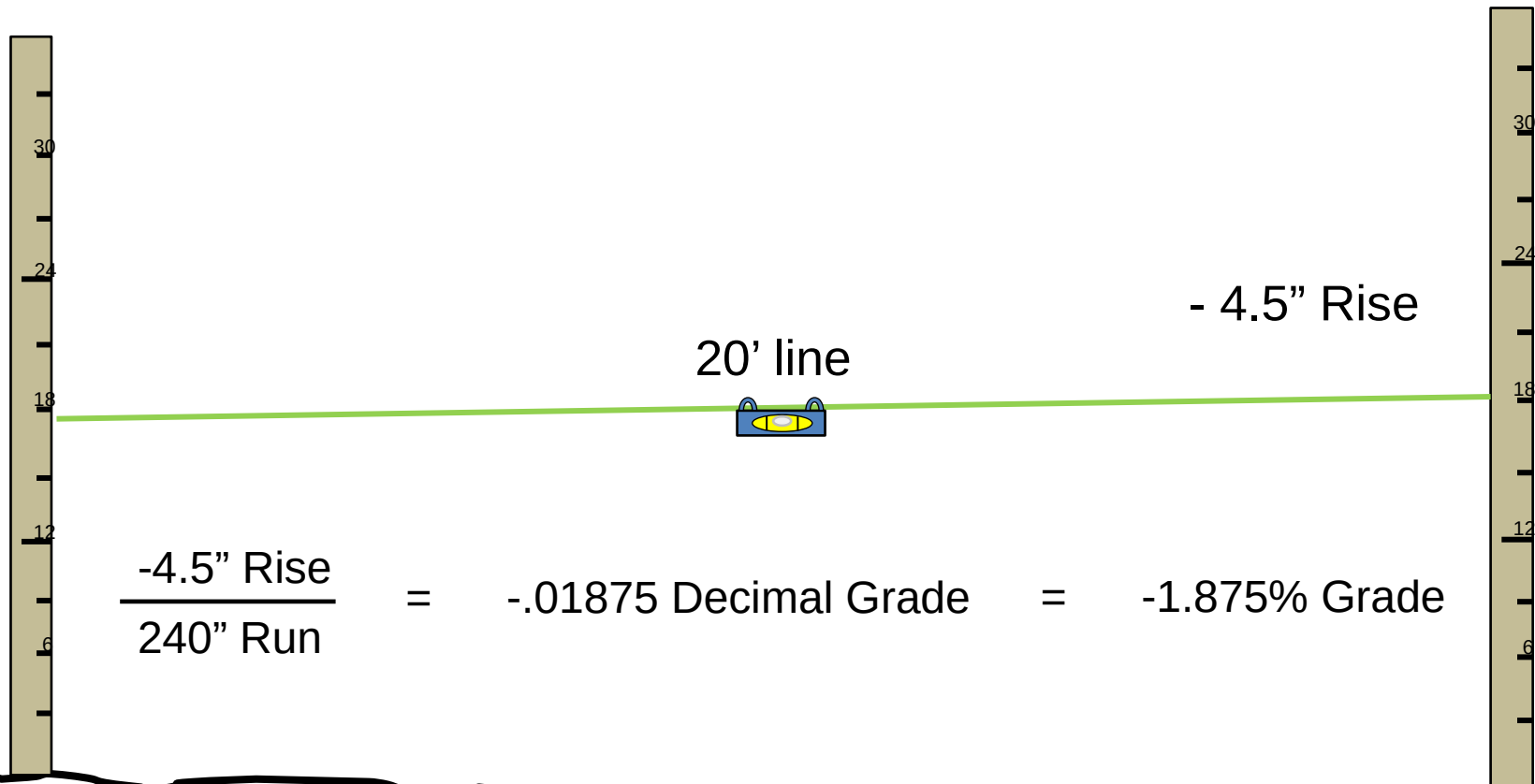
- Line Leveling





Recording Gradients

- Line Leveling





Recording Gradients

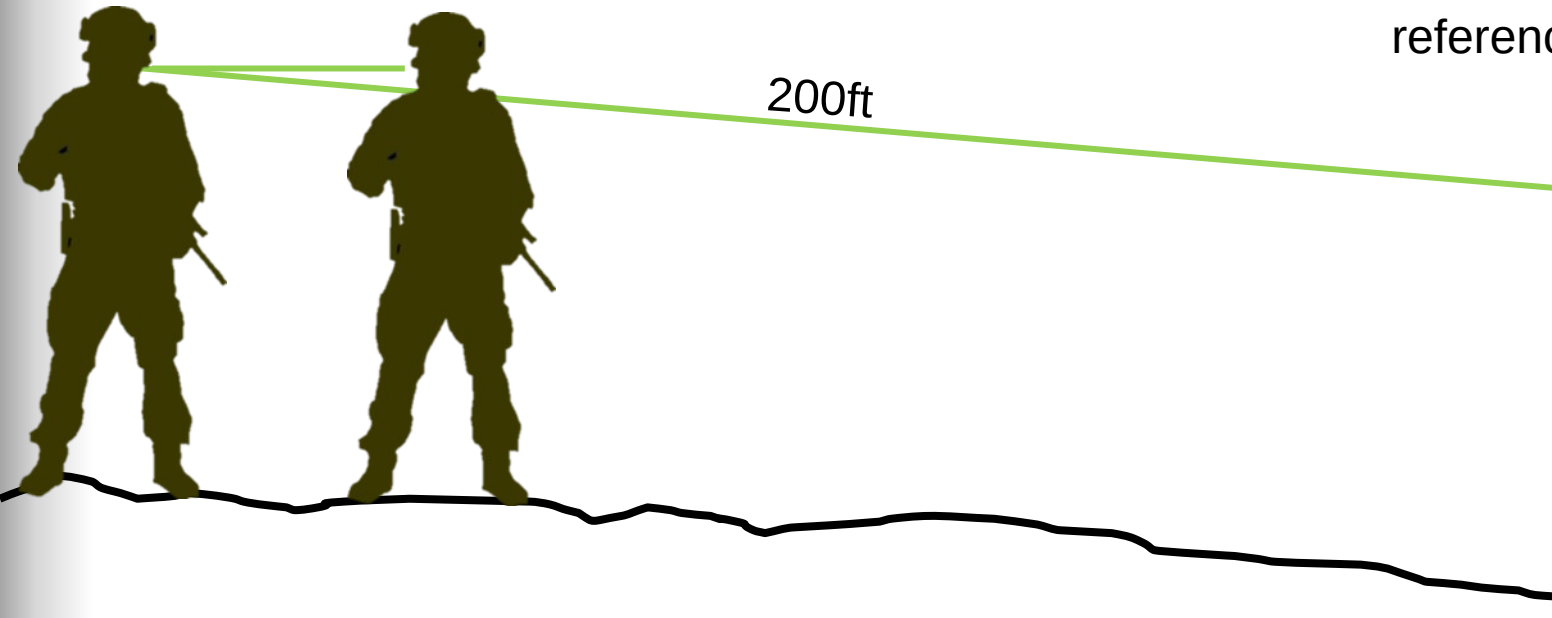
- Clinometer % grade (right side)

Remember the distance between you and your partner. The number indicated will show a positive or negative % grade between you and your partner.

Find "zero%"
reference point

Find "zero %"
reference point

200ft





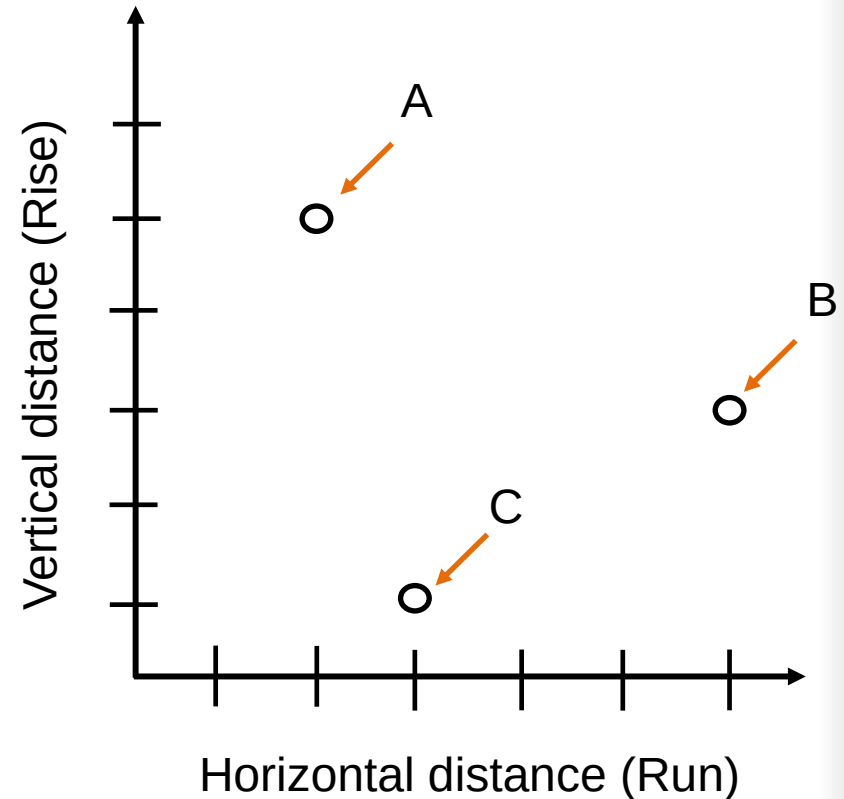
Recording Gradients

Distance	Rise	Decimal Grade	Percent Grade
1-20	-4.5"	-.01875	-1.875%
21-40	-2"	-.00833	-.833%
41-60	-1"	-.0041	-.41%
61-80	+.5"	+.002	+.2%
81-100	+1.25"	+.0052	+.52%
101-120	-1.75"	-.0072	-.72%
121-140	-.5"	-.002	-.2%
141-160	-2.25"	-.0093	-.93%
161-180	-3.5"	-.0145	-1.45%
181-200	+.25"	-.001	-.1%
Cumulative	-13.5"	-.0056	-.56%



Glideslope Ratio

- Glideslope Ratio
 - The relationship of the run to the rise obtained at that distance
 - Divide rise into the run to equal the glideslope ratio



$$\frac{\text{Run}}{\text{Rise}} = \frac{5}{2} = 2.5 : 1$$



Glideslope

- Example: Standing at the end of the AZ, the surveyor is 6000' away from a 2000' Antenna

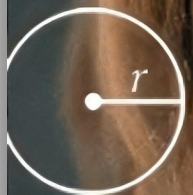
$$\frac{\text{Run}}{\text{Rise}} = \frac{6000'}{2000'} = \frac{3'}{1'}$$

- The glideslope ratio is 3:1, every 3' of run there is a 1' rise



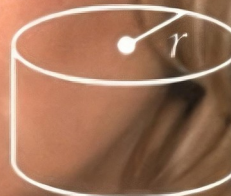
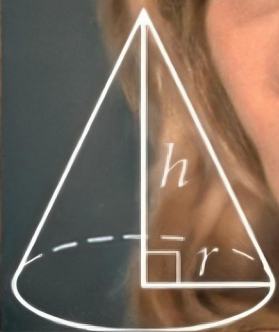
10 MIN BREAK

What's the height of that light pole?



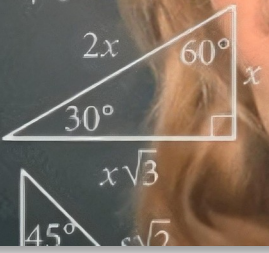
$$A = \pi r^2$$

$$C = 2\pi r$$



$$V = \pi r^2 h$$

	30°	45°	60°
sin	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
tan	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$



$$\int \sin x dx = -\cos x + C$$

$$\int \frac{dx}{\cos^2 x} = \tan x + C$$

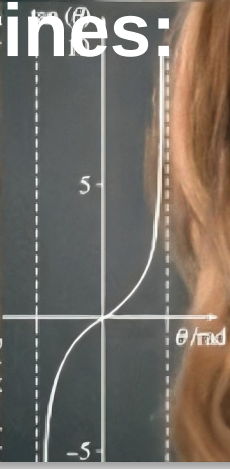
$$\int \tan x dx = -\ln|\cos x| + C$$

$$\int \frac{dx}{\sin x} = \ln\left|\tan \frac{x}{2}\right| + C$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan \frac{x}{a} + C$$

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln\left|\frac{x-a}{x+a}\right| + C$$

Marines:



$$ax^2 + bx + c = 0$$

$$a\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right) = 0$$

$$x^2 + 2\frac{b}{2a}x + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 + \frac{c}{a} = 0$$

$$\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2} = 0$$



Outline

- Mathematics
- **Measurement**
- Review
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Outline

- Mathematics
- Measurement
 - Direction
 - Distance
- Review
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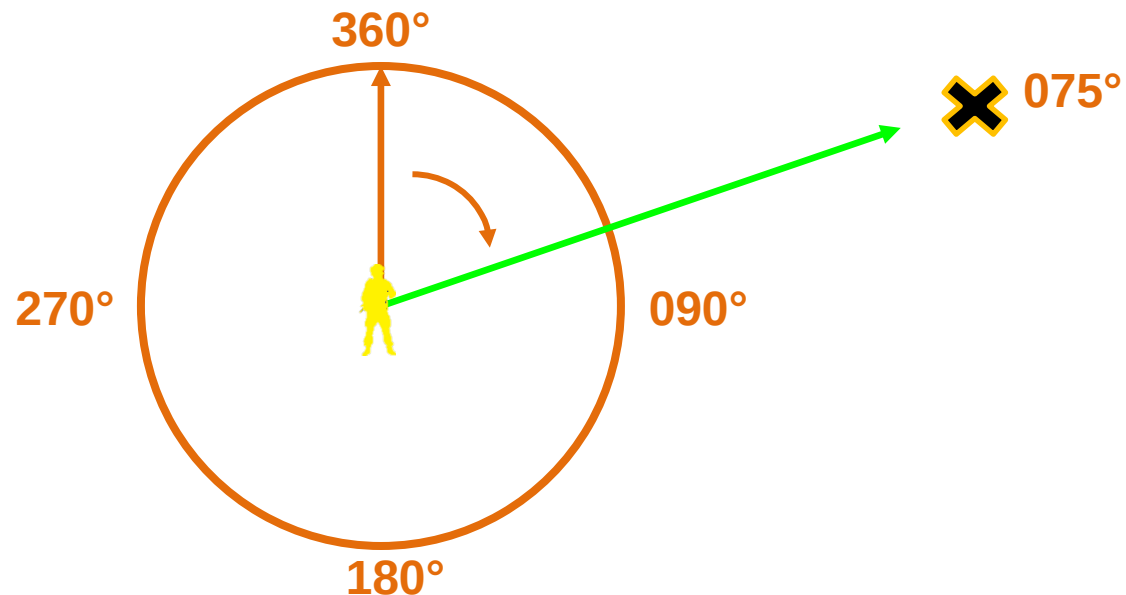


Measuring Direction

- Definitions

- Azimuth

- Horizontal direction observed clockwise from a zero reference line that runs north-south
 - Measure in 1° increments from 001° to 360°

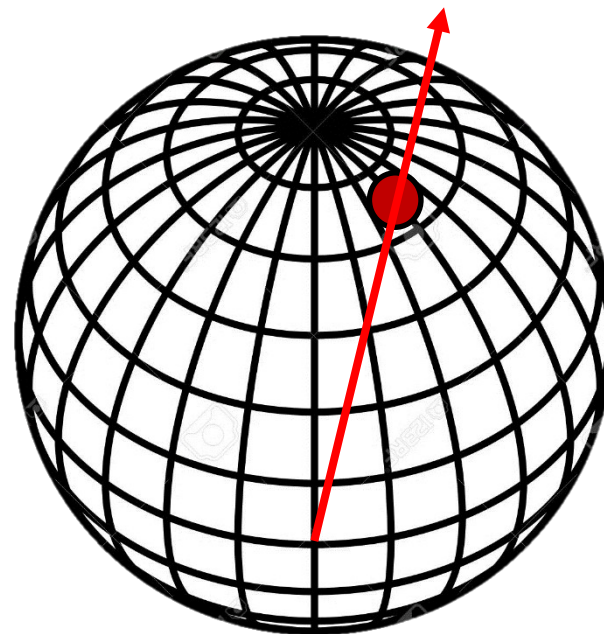




Measuring Direction



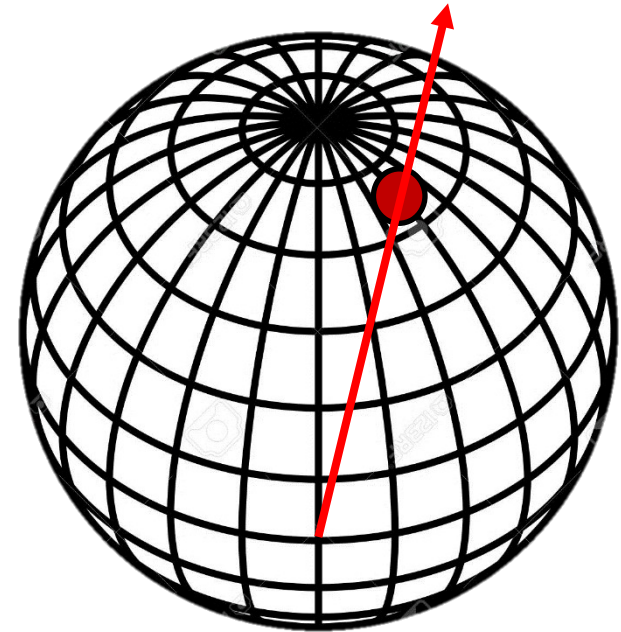
- Definitions
 - True North
 - Based on alignment to Polaris
 - Grid North
 - Based on alignment to the Central Meridian (MGRS)
 - Magnetic North
 - Based on alignment to the Earth's magnetic pole
 - Primary reference used for SAAZA by MMT





Measuring Direction

- Variations in magnetic declination
 - The difference between the True Meridian and Magnetic Meridian is called magnetic declination
 - In the eastern U.S., the compass needle points west of True North and in the west the needle points east of it





Equipment

- Lensatic Compass
 - Accuracy is to $\pm 2.25^\circ$
 - Standard military issue
- Clinometer
 - Reads 0-360° in both 1° increments on the bottom scale and 5° increments on the top scale
 - Accuracy is $\pm 1^\circ$
 - Requires use of both eyes; one sights on an object while the other reads the scale



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Measuring Distance

- Distance measuring procedures
 - When conducting a survey we should concern ourselves with a 5 NM radius
 - This 5 NM radius should note possible hazards to flight from obstacles on the ground (towers, mountains) and in the air (overlapping air space)
 - Time, equipment, and threat will determine what is measured and the level of accuracy



Measuring Distance

- Distance measuring techniques
 - Odometer Readings
 - Pacing Method
 - Each person should have a number of steps that it takes to travel a certain distance
 - A person can either use each foot or every other foot
 - Requires no equipment
 - Can be used covertly



Measuring Distance

- Distance measuring techniques
 - Estimating Distances
 - No equipment is necessary to determine a distance
 - Accuracy depends on the skill of the surveyor
 - Estimating is quick and covert
 - Visual Yardstick



Measuring Distance

- Measuring Tools
 - Tapes
 - Wheels
 - Lasers



Measuring Distance

- Conversions need to be made while surveying
 - 1 Statute mile (visibility) - 5280', 1609.8 m
 - 1 Nautical mile - 6076', 1852 m
 - 1 Meter - 3.28'
 - Odometer: 1/10 mile - 528'



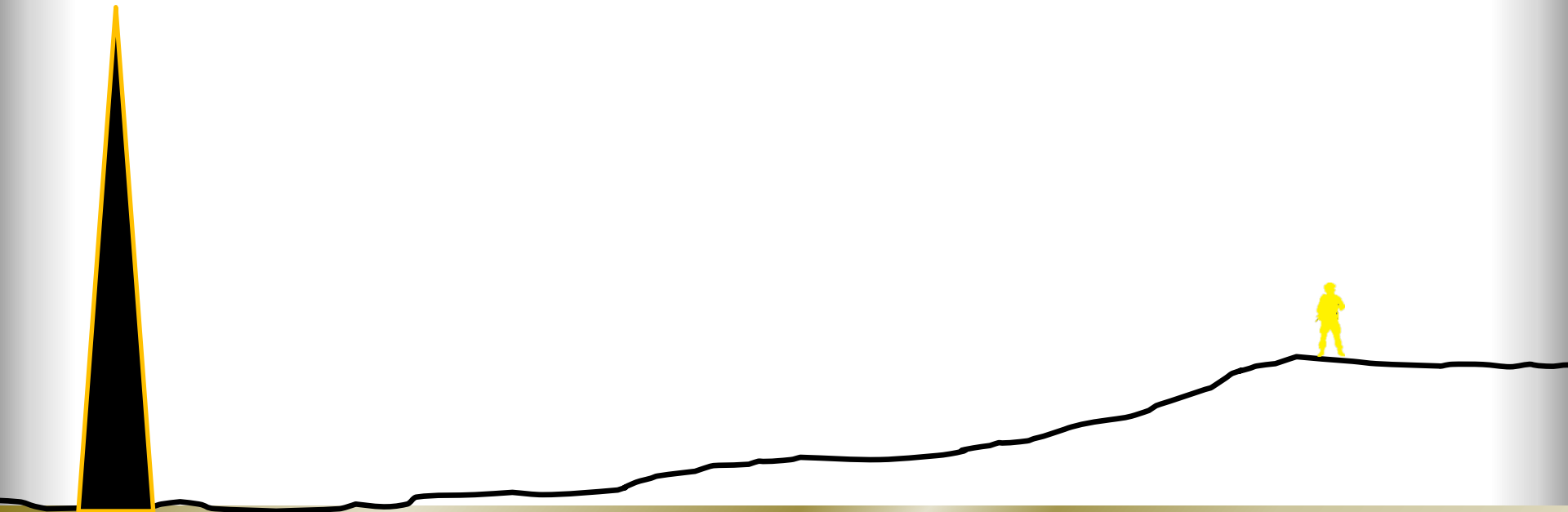
Outline

- Mathematics
- Measurement
- **Review**
- Summary



Review

- Determine angles, distances, and horizontal ranges with trigonometric functions

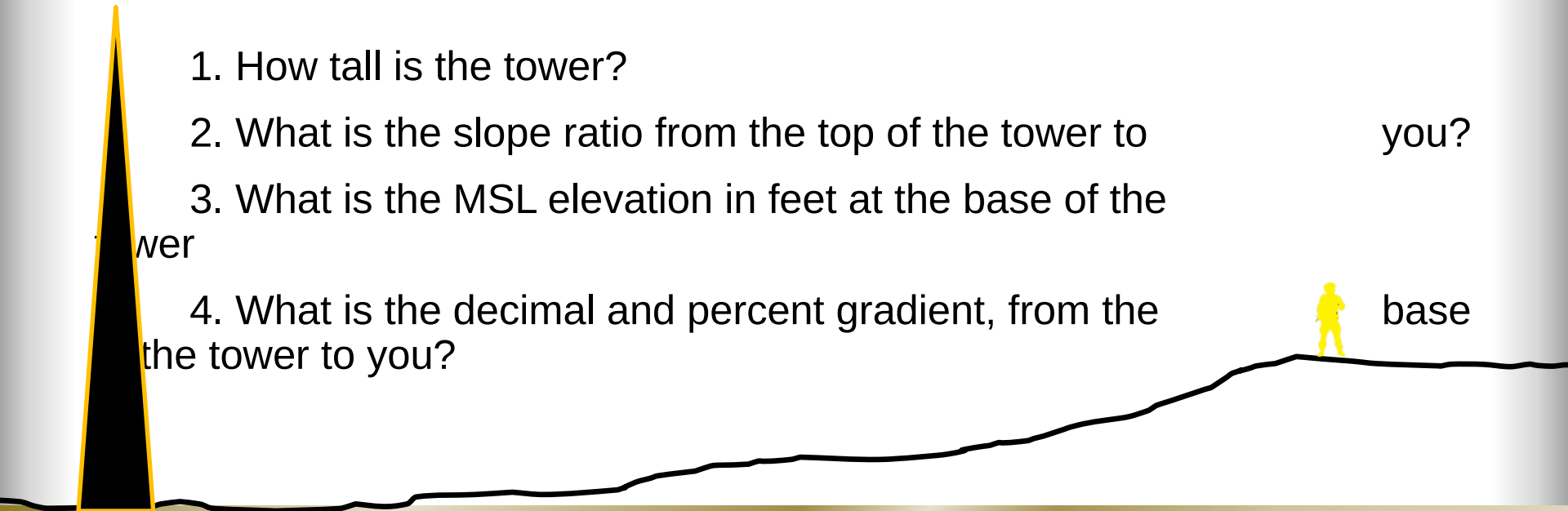




Review

- You are surveying a potential ALZ. Standing at the approach end of the runway 18 overrun, you observe a tall antenna on the 035 azimuth. Your laser range finder reads 3895m from you to the obstacle. Your clinometer reads the angle of elevation to the top of the tower as 19° and an angle of depression to the bottom as 8° . Your DAGR reads your elevation as 2175' MSL.
- Answer the following questions:

- How tall is the tower?
- What is the slope ratio from the top of the tower to you?
- What is the MSL elevation in feet at the base of the tower?
- What is the decimal and percent gradient, from the base of the tower to you?





Review

1. How tall is the tower?

$$\frac{O}{S | H}$$

$$\frac{A}{C | H}$$

$$\frac{O}{19^\circ | 3895}$$

$$\tan(19^\circ) \times 3895 = \text{Opp 1}$$

$$.3443 \times 3895 = \text{Opp 1}$$

$$1341.16 = \text{Opp 1}$$

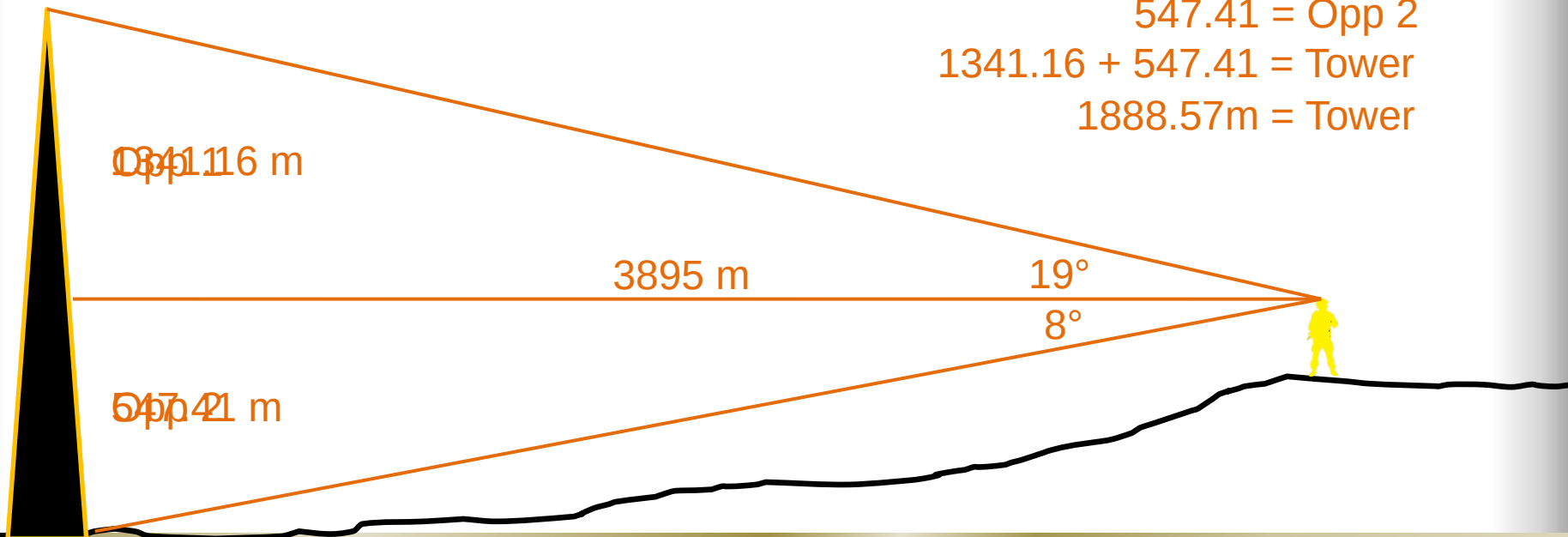
$$\tan(8^\circ) \times 3895 = \text{Opp 2}$$

$$.1405 \times 3895 = \text{Opp 2}$$

$$547.41 = \text{Opp 2}$$

$$1341.16 + 547.41 = \text{Tower}$$

$$1888.57\text{m} = \text{Tower}$$





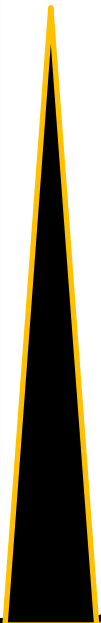
Review

1. How tall is the tower?

1888.57m

2. Converted to feet?

~ 6194.51'





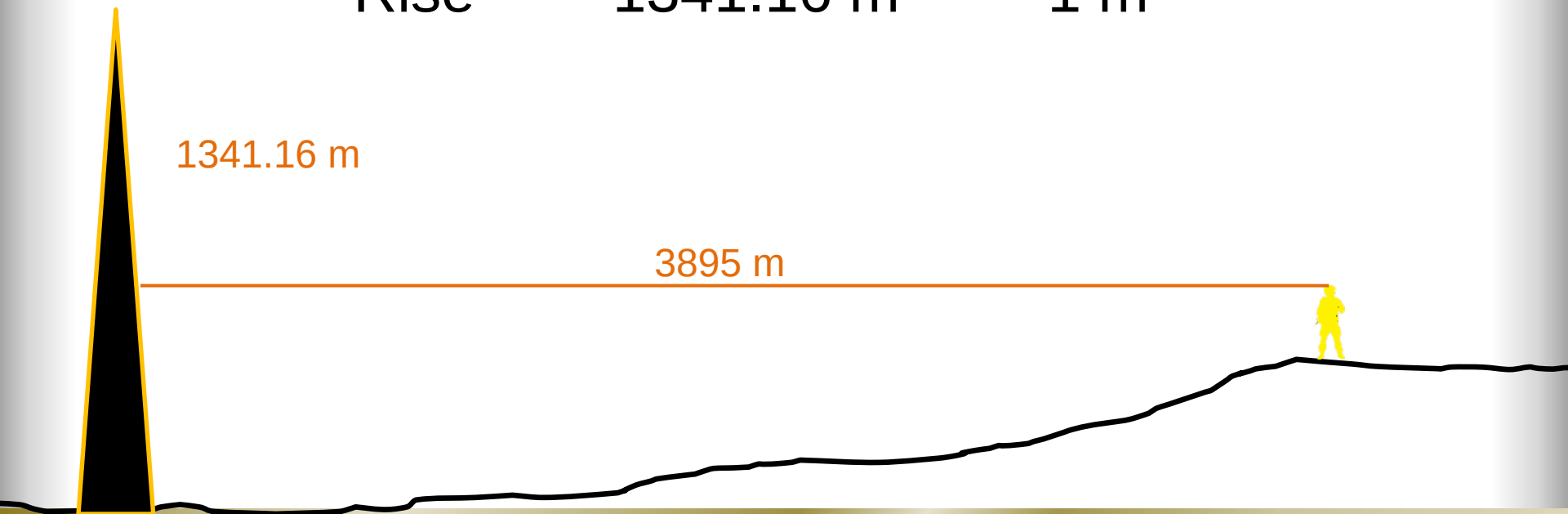
Review

2. What is the slope ratio from the top of the tower to you?

$$\frac{\text{Run}}{\text{Rise}}$$

$$\frac{3895 \text{ m}}{1341.16 \text{ m}}$$

$$\frac{2.9 \text{ m}}{1 \text{ m}}$$

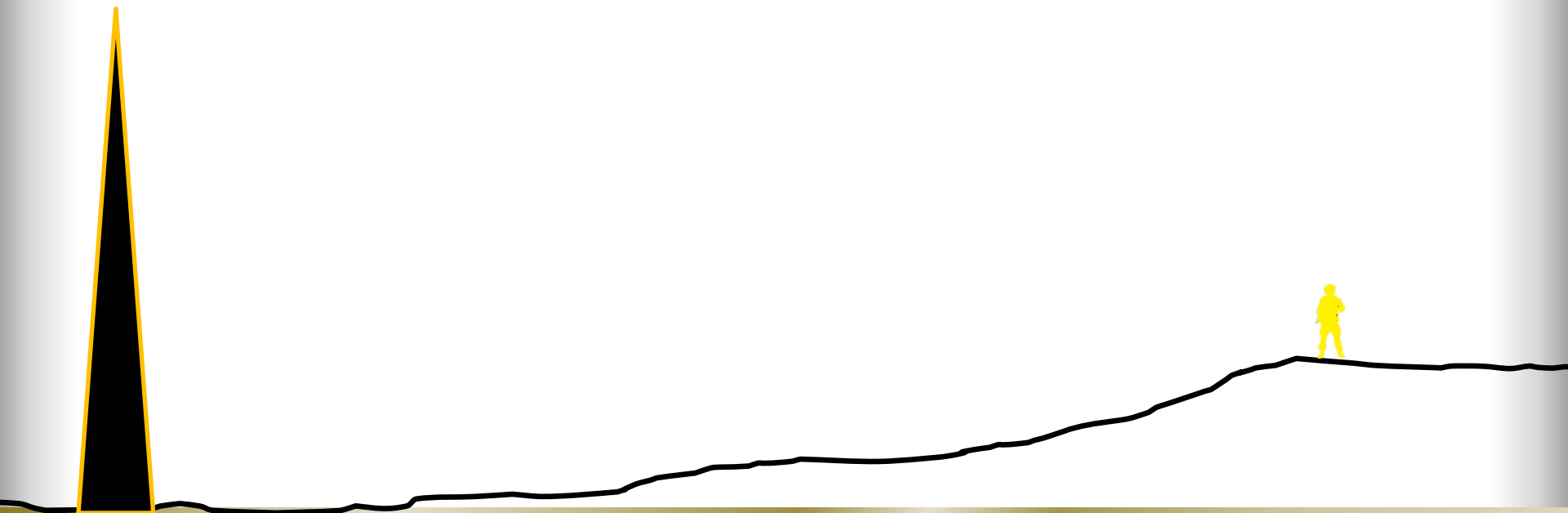




Review

2. What is the slope ratio from the top of the tower to you?

2.9 : 1





Review

3. What is the MSL elevation in feet at the base of the tower?

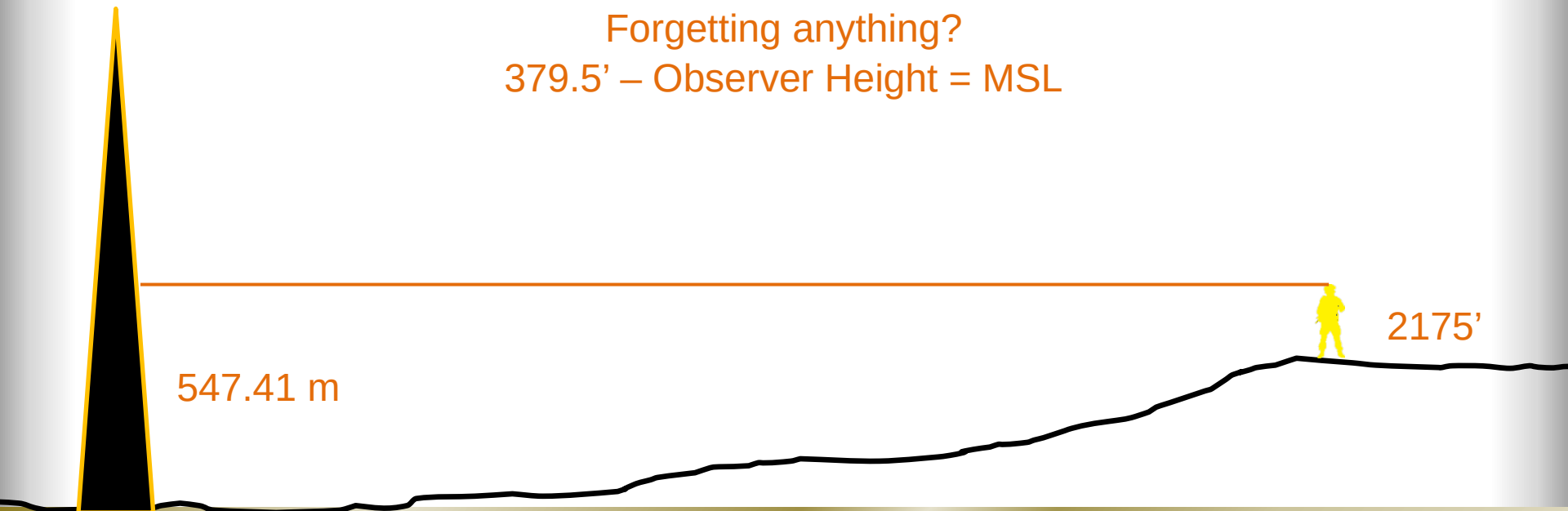
$547.41 \text{ m} \times 3.28 = \text{m to ft conversion}$

$1795.5'$

$2175' - 1795.5' = 379.5'$

Forgetting anything?

$379.5' - \text{Observer Height} = \text{MSL}$

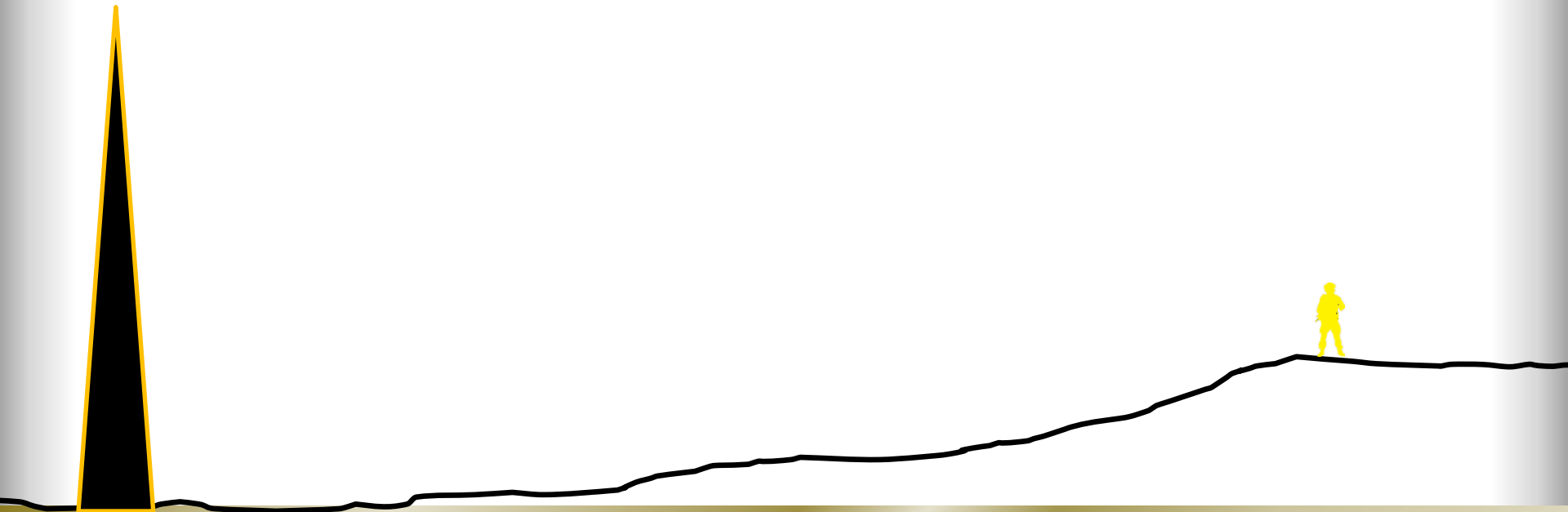




Review

3. What is the MSL elevation in feet at the base of the tower?

379.5' – Observer Height

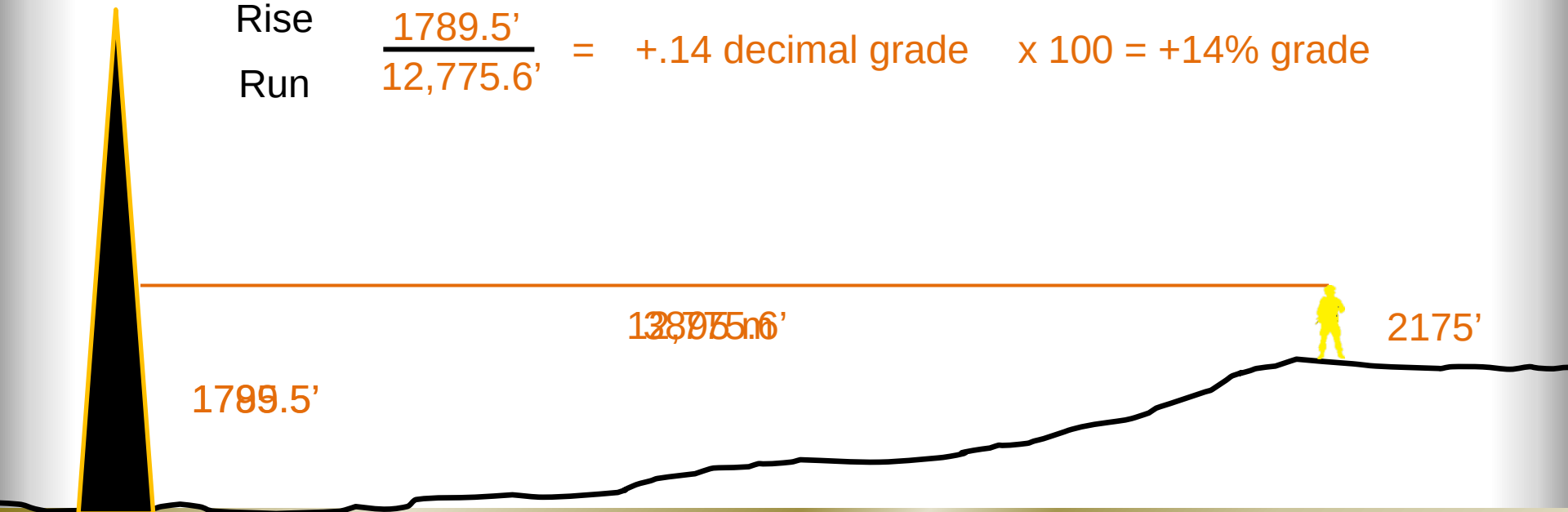




Review

4. What is the decimal and percent gradient, from the base of the tower to you?

$$\begin{array}{l} \text{Rise} \\ \text{Run} \end{array} \quad \frac{1789.5'}{12,775.6'} = +.14 \text{ decimal grade} \quad \times 100 = +14\% \text{ grade}$$





Review

- You are surveying a potential ALZ. Standing at the approach end of the runway 18 overrun, you observe a tall antenna on the 012 azimuth. Your laser range finder read 3895m from you to the obstacle. Your clinometer read the angle of elevation to the top of the tower as 19° and an angle of depression to the bottom as 8° . Your DAGR reads your elevation as 2175' MSL. Answer the following questions:

1. How tall is the tower?

1888.57m / 6194.51'

2. What is the slope ratio from the top of the tower to you?

2.9 : 1

3. What is the MSL elevation in feet at the base of the tower?

380'

4. What is the decimal and percent gradient, from the base of the tower to you?

.1405 / 14.05%





Review

- Identify units of measurement used when conducting SAAZA
 - Unit used for measuring visibility
 - Statute Miles
 - Units used for measuring distances
 - Feet
 - Meters
 - Nautical Miles



Review

- Convert horizontal units of measurement
 - How many feet is the tower from your position?
 - 12,775.6'
 - How many Statute Miles is it?
 - 2.41 SM
 - How many Nautical Miles is it?
 - 2.10 NM
 - What is your elevation in meters?
 - 663.11 m



Outline

- Mathematics
- Measurement
- Review
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Summary

- Mathematics
- Measurement

Questions / Critiques



This Presentation is Classified
UNCLASSIFIED